

Imesha Rajapaksha

Statistics and Operations Research

Operations Research · Optimization · Statistical Modelling · Biostatistics

Sri Lanka | imesharajapaksha2020@gmail.com | +94 78 249 2665

[LinkedIn](#) | [GitHub](#) | [Google Scholar](#) | [ResearchGate](#)

Academic Profile

Statistics and Operations Research graduate with a strong academic foundation in operations research, optimization, mathematical modelling, statistics, and quantitative decision-making, complemented by research experience in biostatistics, epidemiology, clinical research, and healthcare data analysis. Primary academic interests lie in applying operations research and statistical methodology to complex real-world decision problems, particularly through optimization, queueing theory, simulation, network modelling, and predictive analytics, alongside research experience applying quantitative methods to medical and public-health problems, including oncology, post-operative outcomes, and disease surveillance. Currently a Temporary Lecturer in the Department of Community Medicine, Faculty of Medicine, University of Peradeniya, contributing to research involving statistical analysis, research methodology, epidemiology, and health-related data. Seeking MSc/PhD study in Operations Research, Statistics, Biostatistics, Applied Mathematics, or related quantitative disciplines.

Research Interests

- **Operations Research:** Optimization, mathematical modelling, queueing theory, simulation, network optimization, transportation systems, decision modelling, and resource allocation.
- **Statistics:** Statistical modelling, statistical inference, regression analysis, multivariate statistics, experimental design, and quantitative data analysis.
- **Biostatistics:** Clinical data analysis, epidemiology, survival analysis, predictive modelling, medical statistics, and evidence-based healthcare research.
- **Data Science & Machine Learning:** Predictive modelling, machine learning, data preprocessing, statistical learning, and computational methods.

Education

B.Sc. (Hons) in Statistics and Operations Research

2020 – 2025

Faculty of Science, University of Peradeniya, Sri Lanka

Second Class Upper Division; GPA: 3.4/4.00

- Major academic areas: Operations Research, Statistics, Mathematics, and Computer Science.
- Undergraduate training included mathematical methods, optimization, statistical modelling, probability, statistical inference, and computational techniques.

G.C.E. Advanced Level – Physical Science Stream

Bandaranayake Central College, Veyangoda, Sri Lanka

Research Experience

Temporary Lecturer

2026 – Present

Department of Community Medicine, Faculty of Medicine, University of Peradeniya

- Support undergraduate research activities from research design and data collection through statistical analysis and interpretation.
- Contribute to quantitative research involving epidemiological and public-health data.
- Work with Weekly Epidemiological Reports and disease surveillance data for statistical analysis and predictive modelling.

- Apply statistical methods to support evidence-based interpretation of health and disease patterns.

Research Assistant

2026

Department of Surgery, Faculty of Medicine, University of Kelaniya

- Contribute to clinical research involving breast cancer, colorectal cancer, clinicopathological characteristics, and treatment outcomes.
- Perform data cleaning, statistical analysis, interpretation, and preparation of research outputs.
- Apply statistical methods to investigate clinical characteristics, treatment outcomes, recurrence, and prognostic factors.

Research Assistant

2026

Department of Anaesthesiology, Faculty of Medicine, University of Peradeniya

- Contributed to quantitative research and clinical data analysis.
- Assisted with research data management, statistical analysis, and interpretation of healthcare-related findings.

Selected Research Projects

Operations Research, Optimization & Transportation

- **A Multi-Mode Queueing-Based Adaptive Traffic Signal Control Framework for Railway Level Crossings.** Developed a queueing-based adaptive traffic signal control framework for a railway level crossing and three-road intersection, incorporating Poisson arrival processes, queueing theory, mathematical optimization, and simulation using Python and PuLP. The framework considered normal, railway preemption, and recovery operating modes to improve traffic flow and reduce queueing delay.
- **Optimization Mathematical Modelling Using Excel.** Developed optimization models using mathematical programming techniques and Excel-based computational methods for decision-making problems.

Statistics, Biostatistics & Data Science

- **Utility of Modified Glasgow Prognostic Score (mGPS) in Predicting Post-Operative Outcomes in Hepatobiliary and Pancreatic Surgical Patients in Sri Lanka.** Prospective clinical research investigating the utility of mGPS for predicting post-operative outcomes, involving statistical analysis of pre-operative inflammatory and nutritional markers, post-operative complications, length of hospital stay, ICU admission, and mortality-related outcomes.
- **Epidemiology and Clinicopathological Characteristics of Breast Cancer in Sri Lankan Women Under 40 Years of Age: A Registry-Based Study at a Dedicated Breast Centre.** Registry-based study investigating demographic, clinicopathological, receptor-status, lymph-node, tumour-grade, and prognostic characteristics of breast cancer among women younger than 40 years.
- **Breast Cancer Beyond Hormone Receptor Positivity: Clinicopathological Differences Among ER/PR Subtypes.** Evaluated tumour characteristics and clinical outcomes across four hormone receptor (HR) subtypes – ER–/PR–, ER+/PR–, ER+/PR+, and ER–/PR+ – using data from 587 breast cancer patients.
- **Impact of Interval Between Breast Cancer Surgery and Radiotherapy on In-Breast Tumour Recurrence: A Retrospective Single-Centre Study from Colombo North Teaching Hospital, Sri Lanka.** Examined whether the time interval between breast-conserving surgery and adjuvant radiotherapy affects in-breast tumour recurrence, using data from the Colombo North Teaching Hospital breast cancer database (2019–2024); contributed to the statistical analysis.
- **Weekly Epidemiological Report Disease Surveillance and Prediction Dataset.** Developing a structured longitudinal dataset from Weekly Epidemiological Reports covering 13 diseases, 24 districts, and the period 2008–2026, supporting statistical analysis and predictive modelling of

disease patterns.

- **Exploring Heart Disease Predictors with Multivariate Techniques.** Applied Principal Component Analysis, Factor Analysis, Discriminant Analysis, and Canonical Analysis to investigate relationships among heart disease risk factors using multivariate statistical techniques.
- **An Overview of Bayesian Exploration and Modeling of Heart Disease Risk Factors.** Explored Bayesian statistical concepts and modelling approaches for analysing and interpreting heart disease risk factors.
- **Laptop Price Prediction Using Machine Learning.** Developed a web-based machine-learning application using Python, Pandas, NumPy, HTML/CSS, and Random Forest Regression on a dataset of 1,303 observations, achieving approximately 75% prediction accuracy.

Publications

- Published research abstract: **Epidemiology and Clinicopathological Characteristics of Breast Cancer in Sri Lankan Women Under 40 Years of Age: A Registry-Based Study at a Dedicated Breast Centre.** *The Sri Lanka Journal of Surgery, September 2026, Volume 44, Supplement S1, as part of the Sri Lanka Surgical Congress 2026 – 55th Annual Academic Sessions of the College of Surgeons of Sri Lanka.*
- Published research abstract: **Breast Cancer Beyond Hormone Receptor Positivity: Clinicopathological Differences Among ER/PR Subtypes.** *The Sri Lanka Journal of Surgery, September 2026, Volume 44, Supplement S1, as part of the Sri Lanka Surgical Congress 2026 – 55th Annual Academic Sessions of the College of Surgeons of Sri Lanka.*
- Published research abstract: **Impact of Interval Between Breast Cancer Surgery and Radiotherapy on In-Breast Tumour Recurrence: A Retrospective Single-Centre Study from Colombo North Teaching Hospital, Sri Lanka.** *The Sri Lanka Journal of Surgery, September 2026, Volume 44, Supplement S1, as part of the Sri Lanka Surgical Congress 2026 – 55th Annual Academic Sessions of the College of Surgeons of Sri Lanka.*

Conference Presentations

Oral Presentation – SICET 2026 (Paper ID 76) 2026

“A Multi-Mode Queueing-Based Adaptive Traffic Signal Control Framework for Railway Level Crossings.”

Oral Presentation – iPURSE 2026 (Track 1, Paper ID 578) 2026

“Utility of Modified Glasgow Prognostic Score (mGPS) in Predicting Post-Operative Outcomes in Hepatobiliary and Pancreatic Surgical Patients in Sri Lanka.”

Oral Presentation – SURS 2026 (Undergraduate Research Symposium at the University of Peradeniya) 2026

“A Multi-Mode Queueing-Based Adaptive Traffic Signal Control Framework for Railway Level Crossings. Case study of Mirigama, Sri Lanka”

Teaching Experience

Temporary Lecturer 2026 – Present

Department of Community Medicine, Faculty of Medicine, University of Peradeniya

- Undergraduate academic support and research supervision.
- Guidance in research methodology, data analysis, and interpretation.

Seminar Facilitator

Operations Research Society, University of Peradeniya

- Participated in academic seminar and knowledge-sharing activities related to Operations Research and quantitative methods.

Seminar Facilitator

Statistical Circle, University of Peradeniya

- Participated in statistical seminars and academic activities.

Seminar Programme Contributor

Athwela Seminar Programme, University of Peradeniya

Additional Professional Experience

Management Intern – WSAP (NZ) Phase II

2024 – 2025

Career Guidance Unit, University of Peradeniya

- Developed and maintained a database of Statistics and Operations Research students.
- Awarded the WSAP (NZ) Phase II traineeship in October 2024.

Technical Skills

- **Programming & Analysis:** Python, R, RStudio, C, Java
- **Statistical Computing:** Statistical modelling, regression analysis, multivariate analysis, data cleaning, exploratory data analysis
- **Operations Research:** Optimization, mathematical modelling, queueing theory, simulation, network optimization, decision modelling
- **Machine Learning:** Predictive modelling, Random Forest, machine learning workflows, Pandas, NumPy
- **Data Management:** Data preprocessing, database development, structured dataset creation, data quality checking
- **Web Technologies:** HTML, CSS
- **Research:** Research methodology, statistical analysis, clinical data analysis, epidemiological analysis, scientific reporting

Professional Development & Certifications

- Python – Introduction to Data Science and Machine Learning A–Z, Udemy.
- Professional Diploma in Technical Project Management, MTF Institute.
- Business Science Method, Analysis and Research Methodologies, MTF Institute.

Awards & Academic Achievements

- B.Sc. (Hons) in Statistics and Operations Research – Second Class Upper Division, University of Peradeniya.
- WSAP (NZ) Phase II Traineeship – October 2024.

Languages

- Sinhala – Native / Fluent
- English – Professional working proficiency

Academic References

Dr. Nilan Manoj Chathuranga

Senior Lecturer

Department of Mathematics

Faculty of Science

University of Peradeniya, Sri Lanka

chathuranga.mudalige@sci.pdn.ac.lk

Dr. Niluka Rodrigo

Senior Lecturer

Department of Mathematics

Faculty of Science

University of Peradeniya, Sri Lanka

nilukar@sci.pdn.ac.lk